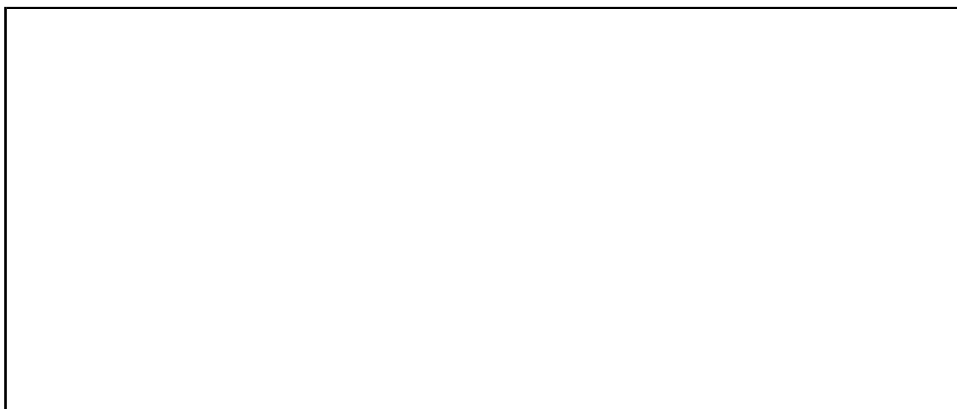


CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Second Semester M.Sc.(Physics) Examination May 2018
PS 717 Mathematical Methods in Physics-II

Date: 02/05/2018, Wednesday **Time:** 01:30 p.m. to 02:00 p.m. **Maximum Marks:** 10

Multiple choice questions



Important Instructions:

- Provide all necessary information as required in the answer sheet.
 - Attempt all questions.
 - Answer the questions in the question paper itself.
 - Use of cell phones is strictly prohibited
 - Use of non-programmable calculators may be allowed if needed.
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Q.1. Choose the correct answer for the following questions. [10]

1. $\int_{|z|=\frac{1}{2}} \frac{dz}{(z+2)(z-1)} = \underline{\hspace{2cm}}.$

- a. $2\pi i$ b. $4\pi i$ c. πi d. 0.

2. A function $f(z) = \bar{z}$ is _____, where $\bar{z}$ is the complex conjugate of z .

- a. nowhere analytic. c. analytic only at origin.
b. an entire function. d. analytic only at $z = 1$.

3. A mapping $w = \frac{1}{z}$ is called _____.
- | | |
|------------------|----------------|
| a. rotation | c. translation |
| b. magnification | d. inversion. |
4. $\int_{|z-1|=2} \frac{dz}{z-1} =$ _____.
- | | | | |
|------|------------|-------------|---------------|
| a. 0 | b. πi | c. $2\pi i$ | d. $3\pi i$. |
|------|------------|-------------|---------------|
5. A function $\frac{e^{\frac{1}{z}}}{z^2}$ has _____ at origin.
- | | |
|--------------------------|----------------------|
| a. removable singularity | c. pole of order two |
| b. essential singularity | d. simple pole |
6. For Legendre polynomials $P_n(x)$, $\int_{-1}^1 P_4(x)P_8(x) dx =$ _____.
- | | | | |
|------------------|------|-------------------|-------|
| a. $\frac{2}{9}$ | b. 0 | c. $\frac{2}{17}$ | d. 1. |
|------------------|------|-------------------|-------|
7. For Bessel function $J_n(x)$, $\frac{d}{dx}(J_0(x))=$ _____.
- | | | | |
|--------------|-------------|-------------|----------------|
| a. $-J_1(x)$ | b. $J_1(x)$ | c. $J_2(x)$ | d. $-J_2(x)$. |
|--------------|-------------|-------------|----------------|
8. For beta function $B(m, n)$, $B(1, 2)=$ _____.
- | | | | |
|------------------|------------------|------------------|-------|
| a. $\frac{1}{6}$ | b. $\frac{1}{4}$ | c. $\frac{1}{2}$ | d. 1. |
|------------------|------------------|------------------|-------|
9. For Hermite polynomials $H_n(x)$, $H_1(50)=$ _____.
- | | | | |
|------|------|-------|---------|
| a. 1 | b. 2 | c. 50 | d. 100. |
|------|------|-------|---------|
10. In tensor analysis, a_{32} represents as_____.
- | | | | |
|-----------------------|-----------------------|-----------------------|-------------------------|
| a. $a_{3j}\delta_2^j$ | b. $a_{2j}\delta_3^j$ | c. $a_{j3}\delta_j^2$ | d. $a_{j2}\delta_3^j$. |
|-----------------------|-----------------------|-----------------------|-------------------------|

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M.Sc.(Physics) Examination May 2018

PS 717 Mathematical Methods in Physics-II

Date: 02/05/2018, Wednesday Time: 02:00 p.m. to 3:30 p.m. Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in same answer sheet.
 2. Make suitable assumptions and draw neat figures wherever required.
 3. Use of non-programmable calculator is allowed.
 4. Show necessary calculations.
 5. Figures to right indicate marks.
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SECTION-I

Q-2 Answer the following questions as directed. [10]

(a) Prove that e^z is an entire function. [02]

(b) For Bessel function $J_p(x)$, prove that $J_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \cos x$. [03]

OR

(b) State generating function for Hermite polynomials $H_n(x)$. Hence prove the recurrence relation $H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x)$. [03]

(c) State Rodrigues formula for Laguerre polynomials $L_n(x)$. Hence find $L_1(x)$ and $L_2(x)$. [03]

OR

(c) State Rodrigues formula for Legendre polynomials $P_n(x)$. Express $5x^2 - 7x + 8$ as a linear combination of Legendre polynomials. [03]

(d) Show that the Kronecker delta is a mixed tensor of order two. [02]

OR

(d) Define contravariant tensor of the second order and dummy index in tensor analysis. [02]

SECTION II

Q-3 Answer the following questions as directed. [15]

(a) Show that $u = \cos x \cosh y$ is harmonic and determine its harmonic conjugate. [03]

(b) Using Cauchy's residue theorem, evaluate the integral $\int_{|z|=1.5} \frac{1-2z}{z(z-1)(z-2)} dz$. [04]

OR

(b) Expand $f(z) = \frac{1}{(z-1)(z+3)}$ in Laurent series valid for (a) $|z| < 1$, [04]
(b) $1 < |z| < 3$ and (c) $|z| > 3$.

(c) Show that the electrostatic potential V between two electric charges which are heads of vectors r and R is given by $V = k \sum_{n=0}^{\infty} \frac{r^n P_n(\cos \theta)}{R^{n+1}}$, where k is constant, θ is the angle between vectors and $P_n(x)$ is Legendre polynomials. [04]

OR

(c) Find the series solution of the Hermite differential equation $\frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + 2ky = 0$, [04]
where k is constant.

(d) Explain the group of symmetries of an equilateral triangle. [04]

OR

(d) Show that $\left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \middle| a, b, c, d \in \mathbb{R}, ad - bc \neq 0 \right\}$ forms a group with multiplication. [04]

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